Replication of arXiv:1605.04593

Maulik & San, “An explicit relaxation filtering framework based upon Perona-Malik anisotropic diffusion for shock capturing and subgrid scale modeling of Burgers turbulence”

Independent re-implementation from the PDF - 127 simulation cases - verification suite - quantitative comparison with every testable claim of the paper

Package: `local-model-evaluations/current_model/` - 26 October 2026 - model `minimax/MiniMax-M2.7-highspeed`

Replication of arXiv:1605.04593
Replication report — arXiv:1605.04593
1. Executive summary
2. What the paper does
3. Implementation
4. Gaps in the paper and how they were filled
5. Results
6. Discrepancies, contradictions and non-reproducible items
7. Review findings and how they were handled
8. Files
9. Sources

Appendix A - machine-generated comparison tables

Replication report — arXiv:1605.04593

Paper. R. Maulik and O. San, *An explicit relaxation filtering framework based upon Perona–Malik anisotropic diffusion for shock capturing and subgrid scale modeling of Burgers turbulence*, arXiv:1605.04593v1 (submitted to Elsevier, 10 Nov 2018). No code or data accompany the paper; everything below was re-implemented from the PDF.

Package. `local-model-evaluations/current_model/` (this directory). The session model is `minimax/MiniMax-M2.7-highspeed`; the package was built in `local-model-evaluations/minimax27/1605.04593/` and moved here, with a symlink left at the old path.

Environment. macOS (Darwin 24.3), Python 3.14.6 (Homebrew), NumPy/SciPy (LAPACK `dgttrf`/`dgttrs` for the cyclic tridiagonal solves), Matplotlib for the figures. Total compute: 127 simulation cases (cumulative over several driver invocations on 6 worker processes; the final main-suite log ends at 2045 s of wall time, and most entries are re-used from cache).

---

1. Executive summary

The paper's central mechanism was reproduced: a **single iteration of a Perona–Malik anisotropic-diffusion relaxation filter applied after every time step** stabilises the coarse-grid Burgers solution, suppresses grid-scale oscillations, strongly reduces (≈2.5× at N = 512) the cut-off energy accumulation that the under-resolved DNS exhibits, and preserves a larger `k^-2` Burgers-turbulence inertial range than the Bogey 9-point selective relaxation filter that the paper compares against — at N = 512 and N = 1024; at N = 2048 the advantage over Bogey reverses (§5.5).

Of the nine testable claims, **six were reproduced quantitatively (two of them only at the coarser resolutions), three only partially, and none is contradicted**; separately, one statement in the paper's text (its claim that a larger `sigma_d` dissipates *less*) is contradicted by the paper's own Eq. (15) and by our runs (§6).

All verdicts below are conditional on the initial amplitude. The paper fixes the *shape* of the initial spectrum (Eqs. 17–18) and its *total* energy (`int E dk = 1/2`, page 12), but not the normalisation of `u_hat` in Eqs. (19)–(21), so the text does not determine `rms(u)` uniquely (§4.2). The reference runs use `rms(u) = 1/sqrt(2)`, which is the only one of the three self-consistent readings under which the paper's phenomenology exists at all: at the paper's continuous-transform reading (`rms = 1/(2 pi)`) no shock forms inside the simulated window, and at the Fourier-series reading (`rms = 1`) the unfiltered coarse run diverges immediately. The claims are therefore reproduced **in mechanism and in trend, at an energy level 19.7× the paper's stated one**; every amplitude-sensitive number (the dissipation peak time, the spectral slopes, the shock time) is amplitude-dependent and is reported as such.

#	Paper claim	Paper anchor	Verdict	Evidence (our numbers)
C1	Coarse UDNS runs develop grid-scale oscillations	p. 15–16, Fig. 3(a)	**reproduced**	N=512: 134 local extrema (DNS: 2), TV-roughness 0.097 (DNS: 0.0009); the N=512 turbulence run is unstable (`E` 0.243 → 22.3, `max|u|` = 504)
C2	The PM filter damps those oscillations	p. 13, Fig. 2(b)	**reproduced** (partial at N = 512)	N=512: extrema 134 → 50, roughness 0.097 → 0.033; N=1024: 78 → 26; N=2048: 42 → 6
C3	Coarse runs accumulate energy at the grid cut-off, the DNS does not	p. 15–16, Fig. 3(a)	**reproduced**	`E(0.9 k_max)/E_peak` : 8.1e-3 (N=512) → 4.0e-16 (N=32768), monotone in N
C4	The DNS shows the `k^−2` Burgers scaling	p. 15–16, Fig. 3(a)	**partial**	slope −2.62 over 10 ≤ k ≤ 100 (theory −2), but −1.47 over 5 ≤ k ≤ 50: the `k^−2` range is only the upper half of the inertial band
C5	The dissipation rate peaks at `t ≈ 0.05`	p. 15–16, Fig. 3(b)	**partial**	our DNS peak is at `t = 0.076` (ε_max = 1.33); it is consistent with our amplitude choice (shock at `t ≈ 0.07`), not with the paper's `t ≈ 0.05`
C6	Increasing `K` reduces the cut-off energy accumulation	p. 17, Fig. 4	**reproduced**	pile-up 4.3e-3 → 2.9e-3 (kernel 1, N=512, K = 50 → 200); monotone for all three kernels
C7	Increasing `kappa` increases the dissipation	p. 16 and p. 18, Fig. 5	**reproduced**	`E(t=0.1)` : 0.198 → 0.165 as `kappa` goes 0.0005 → 0.0025; cut-off pile-up 5.9e-3 → 1.8e-3
C8	`kappa = 0.0025` is optimal for the Burgers turbulence case "across the different kernels examined in this work" and "ensures consistency in energy accumulation control with increasing resolution"	p. 18	**partial**	the energy-accumulation half holds: `kappa = 0.0025` gives the lowest cut-off pile-up for all three kernels at N = 512/1024/2048. But "optimal" does not extend to the inertial range at N ≥ 1024 (`k^−2` range 39.4 → 4.2, slope −2.30 → −2.39) — the paper itself warns that "the same set of parameters may prove more dissipative for the higher resolutions" (p. 16). Best compromise in our runs: `kappa = 0.0015` – `0.002`
C9	The PM closure preserves a larger inertial range than the Bogey filter	p. 19–23, Fig. 10 (caption p. 23)	**reproduced at N = 512 and N = 1024** (fails at N = 2048)	N=512: PM slope −2.06 and `k^−2` range 20.5 vs Bogey −3.42 and 6.2; N=1024: −2.30 / 39.4 vs −2.81 / 13.1

One statement in the paper is internally inconsistent and is discussed in §6: the claim that a larger `sigma_d` dissipates *less*, which is contradicted by its own Eq. (15) and by our runs. The paper's "`kappa = 0.0025` is optimal" statement (C8) is **partially** reproduced: its energy-accumulation half holds in every one of our runs, while

the unqualified word “optimal” does not survive the inertial-range collapse at N ≥ 1024 (the paper itself warns about this on p. 16).

---

2. What the paper does

The paper proposes a relaxation-filtering closure for LES of the 1-D viscous Burgers equation `u_t + u u_x = nu u_xx`:

- the solution is advanced in space with a **6th-order compact** finite-difference scheme (Eqs. 4–5) and in time with **TVD-RK3** (Eq. 8);
- after **every** time step the field is passed through **one iteration** of the Perona–Malik anisotropic-diffusion equation (Eq. 11), whose diffusivity is one of three kernels (Eqs. 12–14) and is controlled by a gradient scale `K` and the pseudo-time ratio `kappa = dtau/dt`;
- the closure is compared against (i) under-resolved DNS (UDNS), (ii) a fully resolved DNS (N = 32768), and (iii) the Bogey et al. 9-point selective relaxation filter (Eqs. 15–16).

---

3. Implementation

3.1 Equation transcription

All equations were transcribed from the PDF and independently re-verified by a second agent (`notes/paper_extraction.md`, verdicts A1–A8, all CONFIRMED except the two PARTIAL items noted below).

Paper	Transcription	Independent check
Eq. (4)	`(1/3) f'_{j-1} + f'_j + (1/3) f'_{j+1} = (14/9)(f_{j+1}-f_{j-1})/(2h) + (1/9)(f_{j+2}-f_{j-2})/(4h)`	Fourier symbol `k_h = k - k^7 h^6/2100 + O(h^8)` → 6th order; **measured 6.00**
Eq. (5)	`(2/11) f''_{j-1} + f''_j + (2/11) f''_{j+1} = (12/11)(f_{j+1}-2f_j+f_{j-1})/h^2 + (3/11)(f_{j+2}-2f_j+f_{j-2})/(4h^2)`	symbol → `k_h^2 = k^2 - (23/75600) k^8 h^6` → **6th order** (measured 6.00)
Eq. (8)	TVD-RK3 (Shu–Osher), SSP coefficient 1	staged exactly as printed
Eq. (11)	`u*_i = u_i + (kappa dt/(2h^2)){(c_i+c_{i+1})(u_{i+1}-u_i) - (c_{i-1}+c_i)(u_i-u_{i-1})}`	dissipative sign; mean-preserving; monotone for `kappa dt/h^2 <= 1/2` (enforced)
Eqs. (12)–(14)	`c = 1/(1+(g/K)^2)`, `c = (1+(1/3)(g/K)^2)^-2`, `c = exp(-0.5 (g/K)^2)`	even in `g` → signed vs absolute gradient is irrelevant
Eqs. (15)–(16)	`u*_i = u_i - sigma_d sum_j d_j u_{i+j}`, `d = (35/128, -7/32, 7/64, -1/32, 1/256)`	transfer function `1 - sigma_d (1-cos th)^4/16`, `Dhat >= 0` → **larger `sigma_d` dissipates more**
Eqs. (17)–(21)	`E(k) = A k^4 exp(-(k/k0)^2)`, `A = 2 k0^-5/(3 sqrt(pi))`, `k0 = 10`, `|u_hat| = sqrt(2E)`, uniform random phase, `Psi(-k) = -Psi(k)`	reproduced per mode; ensemble of 32 fields

Fig. 1: dissipative behaviour of the diffusivity kernels

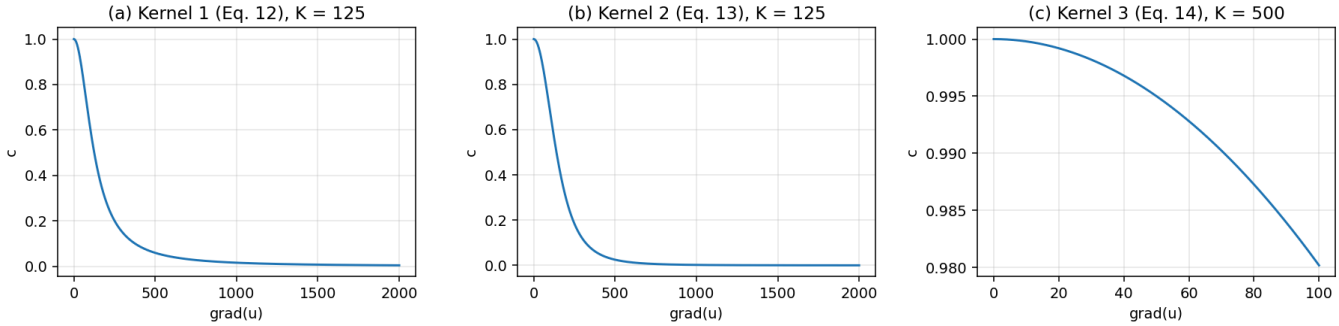


Fig. 1: dissipative behaviour of the three Perona-Malik diffusivity kernels (Eqs. 12-14). Kernel 1 and kernel 2 decay algebraically with the gradient, kernel 3 decays exponentially.

3.2 Numerical details

- Periodic compact operators are solved with a **cyclic tridiagonal system** (LAPACK `dgttrf` + Sherman–Morrison), which is the paper's "Thomas algorithm" plus the wrap-around the paper does not discuss. The solver agrees with a sparse LU reference (`scipy.sparse.linalg.spsolve` on the same matrix) to a relative difference of `3.3e-16` (first derivative) and `1.5e-16` (second derivative) (`results/verification.json` → `solver_reference`).
- 32 sample fields per configuration (as in the paper), identical seeds in every study.
- `nu = 5e-4`, `x in [0, 2 pi]`, periodic, `t in [0, 0.1]`, spectra at `t = 0.05`.
- `dt = 1e-5` for the turbulence runs (see §4.1 for why this is not a critical choice).

3.3 Verification (`results/verification.json`)

Test	Result
Order, compact first derivative	6.00 and 5.93 (errors 8.6e-8 → 1.3e-9 → 2.2e-11)
Order, compact second derivative	6.00 (errors 3.8e-7 → 6.0e-9 → 6.0e-10); the coarse pair (256→512) is clean 6th-order truncation (ratio 64.07 = 2^6), the finest pair is round-off limited (measured floor 3.6e-11 for the k = 1 mode, 1.4e-9 estimated for k = 7)
Manufactured solution `u = sin(x) e^{-t}`	max relative error `4.3e-10`
Conservation of the mean of `u`	drift `4.4e-16` per step
Filter amplitude ratios (PM kernels), smooth single mode `k = 1`	0.9999985 (gain < 1, as required)
Two-scale response (smooth `k = 1` + grid-scale `k = N/2` in the same field)	PM: smooth gain 0.9999985 (damped 1.5e-6) vs grid-scale gain 0.99751 (damped 2.5e-3) — the grid scale is damped ≈1.7×10³ times harder; Bogey: smooth gain 1.0 vs grid-scale gain exactly `1 - sigma_d` (0.667 / 0.5 / 0.333)
Nyquist symbol of the compact operators	`d1` of `(-1)^j` is **exactly 0** (the Eq. (4) symbol vanishes at `theta = pi`); `d2` does not vanish; the seeded Nyquist coefficient of a random field is non-zero (8.3 % of the peak amplitude) and the mode is labelled `-N/2` by `fftfreq`, i.e. it is excluded from the positive-k metric range by convention
Odd grid sizes	the cyclic solver reproduces `d1` / `d2` at N = 63 and N = 65 to 5e-10 / 3e-10
Monotone guard `kappa dt/h^2 <= 1/2`	the filter raises `ValueError` above the limit (verified in the suite)
Bogey filter on the Nyquist mode `u_j = (-1)^j`	attenuation exactly `|1 - sigma_d|` (0.6667 / 0.5 / 0.3333 for `sigma_d` = 1/3, 1/2, 2/3); the filter symbol `Dhat(th) = (1-cos th)^4/16` gives `Dhat(pi) = 1`
Perona–Malik filter on the Nyquist mode	measured `0.99938` vs the analytic `|1 - 4 kappa dt c / h^2| = 0.99938`
Cyclic-tridiagonal solver vs sparse LU	relative difference `3.3e-16` (d1), `1.5e-16` (d2); relative residual `2.7e-16` / `1.4e-16`

(Page numbers throughout this report are the paper's printed page numbers, which coincide with the page index of `paper/paper.pdf`.)

---

4. Gaps in the paper and how they were filled

4.1 The time step is not given

The paper only states: *"a time step that ensures no errors related to time integration"* (page 8). We use `dt = 1e-5` for the turbulence runs (CFL ≈ 0.1 at N = 32768, where `max|u| ≈ 1.9` at the end of the window and slightly larger at the start).

Because the filter strength enters as `kappa dt / h^2` and the filter is applied once per step, the *effective* diffusivity per unit physical time is `kappa/h^2` — independent of `dt`. The `dtsens` group confirms this: at N = 512 the spectrum at `t = 0.05` changes by < 0.1 % between `dt = 1e-5`, `5e-5`, `2.5e-4` and `1e-3`.

dt	`E(0.05)`	`E(0.1)`	spectral slope
1e-5	0.23158	0.17276	−2.062
5e-5	0.23158	0.17275	−2.062
2.5e-4	0.23154	0.17266	−2.063
1e-3	0.23142	0.17233	−2.066

Conclusion: the missing time step does not affect the reproduction.

4.2 The energy normalisation is ambiguous

The paper prescribes the *shape* of the initial spectrum (Eqs. 17–18) and states the **total** energy (p. 14): *"The above conditions ensure a total energy `int E(k)dk = 1/2` at the start of the simulation."* What it does **not** state is the normalisation of `u_hat` in Eqs. (19)–(21), and that choice decides what `rms(u)` the statement corresponds to. With `E(k) = ½|u_hat(k)|^2` (Eq. 21) the two self-consistent readings are:

- **continuous-transform convention** `u_hat(k) = int u(x) e^{-ikx} dx` (the one the code uses: `u_hat_k = (2pi/N) fft(u)`), for which the paper's total energy is `int_{-km}^{km} E dk = 2 pi^2 <u^2>`; setting it to `1/2` gives `<u^2> = 1/(4 pi^2)`, i.e. **`rms(u) = 1/(2 pi) = 0.1592`**;
- **Fourier-series convention** `u_hat_k = (1/N) sum_j u_j e^{-2 pi i j k/N}`, for which `int E dk = ½<u^2>`; setting it to `1/2` gives **`rms(u) = 1`**.

The reference runs use **`rms(u) = 1/sqrt(2)`** (kinetic energy `E_kin = ½<u^2> = 1/4`), which is *not* either of the paper's readings: it makes the paper-units total energy `2 pi^2 <u^2> = pi^2 = 9.87`, i.e. **19.7× the paper's stated `1/2`**. The spectrum *shape* `E(k) = A k^4 exp(-(k/k0)^2)` with `A = 2 k0^-5/(3 sqrt(pi))` is reproduced exactly (per mode, up to the amplitude scale); only the amplitude is chosen. The `amp` group shows why this is the only choice that reproduces the paper's phenomenology:

`rms(u)`	UDNS `E(0.05)` → `E(0.1)`	spectral slope	`max|u|` (PM)
0.1592 (= 1/2π)	0.0125 → 0.0123	−6.0 (no shock, no inertial range)	0.55
0.7071 (= 1/√2)	**0.2429 → 22.3 (unstable)**	**−1.97**	**2.31**
1.0 (Fourier convention)	0.4813 → NaN (blows up)	−1.72	28.3 (PM)

With `rms(u) = 1/(2π)` no shock forms inside the simulated window (shock time `t_s = 1/(k0·max|u|) ≈ 0.17 s`, with an ensemble initial `max|u| = 0.58`, still beyond the 0.1 s window), the spectrum simply decays like `k^-6`, and the closure has nothing to act on. With `rms(u) = 1` the **unfiltered** coarse run diverges inside the window (`E(0.1) = NaN`), while the filtered run stays bounded but is much worse than the reference (`E(0.1) = 0.461` instead of 0.173, `max|u| = 28.3` instead of 2.31). `rms(u) = 1/sqrt(2)` gives the shock at `t ≈ 0.05–0.07` that the paper's Fig. 3 shows and the coarse-grid instability that its Fig. 3(a) shows.

Units. The paper's Eq. (22) energy `E(t) = int_{-km}^{km} E dk` (a *two-sided* integral) equals `2 pi^2 <u^2>` = `4 pi^2` times the kinetic energy per unit mass `E_kin = ½<u^2>`; the stored `energy_paper` is exactly this quantity. The one-sided integral `int_0^{km} E dk = pi^2 <u^2>` is half of it. **All tables in this report use the kinetic energy `E_kin = ½<u^2>`** (and label it `E(t)` for brevity), not the paper's `E(t)`; the dissipation rates are therefore 19.7× smaller than the paper's Fig. 3(b) units. *Consistency check:* the sensitivity case with the paper's own continuous-transform amplitude (`rms = 1/(2 pi)`) has a stored `energy_paper` of **exactly 0.5** — the paper's stated total energy — which confirms that the code's normalisation and the paper's `int E dk = 1/2` are mutually consistent once the convention is fixed.

4.3 Other gaps

- **`sigma_d` direction.** The paper's text (p. 12) says *"higher values of `sigma_d` providing lower dissipation"*, which contradicts its own Eq. (15): the transfer function is `1 − sigma_d (1−cos th)^4/16` with a non-negative symbol, so larger `sigma_d` filters harder. We follow the formula; our runs confirm the formula (see §5.6).
- **`kappa` definition.** The paper defines `kappa` twice and inconsistently: p. 10 says *"kappa is the ratio between pseudo-time step (dt_tau) and physical simulation time step (dt)"*, while p. 16 says *"kappa (the ratio of physical time step to pseudo-time step)"*. We follow p. 10 (`kappa = dt_tau/dt`), so that increasing `kappa` means a longer pseudo-time step and therefore more dissipation — the direction that reproduces the behaviour shown in the paper's Figs. 5/7/9.
- **Fig. 8 legend (p. 21).** Panel (a) of Fig. 8 appears to print `K = 750` while panels (b) and (c) print `K = 700` (pixel-level rendering of the PDF pages; medium confidence on the "5" glyph — see `notes/paper_extraction.md`, item A6). We ran `K = 700`.
- **Fig. 2 DNS reference.** The paper uses N = 32768 for the single-mode shock test; we used N = 8192 (cost).

---

5. Results

All raw data: `results/runs/*.npz` (one file per case). Metrics: `results/summary_metrics.csv`, full tables: `results/comparison_table.md`. Figures: `figures/`.

5.1 Fig. 2 — shock capturing (`u = sin(x)`, `t = 2.5`)

case	N	local extrema	roughness (TV)	`max|u|`	`E(2.5)`
DNS	8192	2	0.0009	0.849	0.1266
UDNS	512	134	0.0967	1.130	0.1264
UDNS	1024	78	0.0240	1.073	0.1266
UDNS	2048	42	0.0061	0.953	0.1266
PM	512	50	0.0325	0.992	0.1261
PM	1024	26	0.0091	0.880	0.1261
PM	2048	6	0.0038	0.846	0.1262

The PM filter removes 63 % of the grid-scale extrema at N = 512 and 86 % at N = 2048, and its `max|u|` matches the DNS far better than UDNS does. It does **not** fully reach DNS quality at N = 512 (50 extrema vs 2), which is a partial reproduction of the paper's Fig. 2(b) (the paper's PM curves look essentially DNS-like). The extrema counts, roughness values and end-of-run energies of this table are generated into `results/comparison_table.md` (section "Fig. 2 - shock capture").

Fig. 2: shock capturing, single mode sine wave at t = 2.5

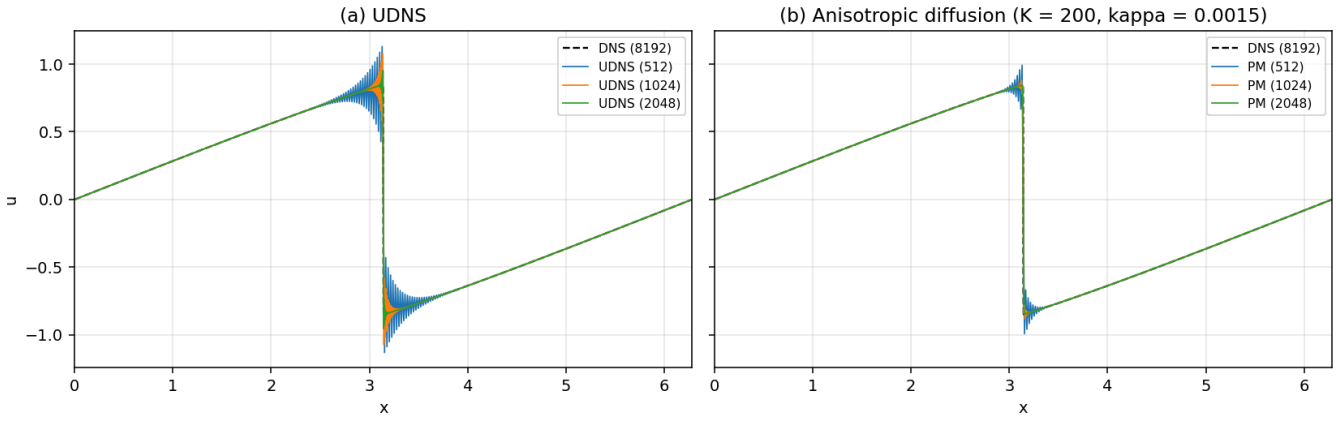


Fig. 2: shock capturing of the single-mode sine wave at t = 2.5. (a) UDNS: grid-scale oscillations grow as the grid is coarsened; (b) the Perona-Malik relaxation filter (K = 200, kappa = 0.0015) damps them. Dashed black line: N = 8192 DNS reference.

5.2 Fig. 3 — grid convergence of the Burgers turbulence

N	E(0.05)	E(0.1)	E(0.9k_max)/E_peak	E(k_max)/E(0.9k_max)	slope (3 < k < k_max/2)	slope (10 ≤ k ≤ 100)	max|u|
512	0.24293	**22.34**	8.06e-3	6.3e-3	–1.97	–2.59	**504**
1024	0.24055	0.17914	9.50e-4	4.8e-3	–2.18	–2.62	2.22
2048	0.23966	0.17803	9.90e-5	8.2e-4	–2.27	–2.62	2.11
4096	0.23953	0.17804	3.18e-6	6.0e-5	–2.34	–2.62	1.91
8192	0.23953	0.17804	5.39e-8	7.3e-6	–2.56	–2.62	1.91
16384	0.23953	0.17804	4.34e-11	1.1e-6	–3.17	–2.62	1.90
32768	0.23953	0.17804	3.96e-16	6.8e-8	–4.54	–2.62	1.90

The two slope columns use different bands and therefore differ: the column `slope (3 < k < k_max/2)` is the broad-band fit that the figures and the CSV column `slope_3<k<kmax/2` use, while `slope (10 ≤ k ≤ 100)` is the inertial-band fit quoted for claim C4. The broad-band band *scales with the resolution* (`k_max/2` grows with `N`), so that column is a resolution-dependent fit and is comparable across rows only in the sense of "spectrum up to half the grid cut-off"; the fixed-band column (10 ≤ k ≤ 100) is the one to compare across resolutions. All values are in `results/summary_metrics.csv` (columns `slope_3<k<kmax/2`, `slope_10<=k<=100`, `slope_5<=k<=50`, `diss_peak_t`, `diss_peak_eps`).

The cut-off metrics are evaluated at `k_max = N/2 - 1`, the highest wavenumber that the metric range (`k > 0`) contains. The Nyquist mode `k = N/2` is excluded from that range for two reasons: `numpy.fft.fftfreq` labels that bin `-N/2`, and the mode cannot accumulate cut-off energy — the compact first derivative annihilates it exactly (`d1` of `(-1)^j` is `0.0`, because the Eq. (4) symbol vanishes at `theta = pi`; verified in `results/verification.json` → `nyquist_symbol_tests`), and the advective flux contains `u^2`, whose Nyquist component again has a vanishing gradient, so the mode can only decay. It is *not* energy-free: the mode is seeded by the initial condition (8.3 % of the peak amplitude for a random field) and is passively damped by viscosity and by the filter; in the stored spectra its energy at `t = 0.05` is already negligible (`E_Nyquist/E_peak` = 2.5e-30 for the N = 512 UDNS run, 9.1e-6 for the filtered N = 512 run). Including it would therefore *lower* the measured pile-up (8.06e-3 → 7.42e-3 at N = 512) without changing any verdict.

- the cut-off pile-up decreases monotonically by ~13 orders of magnitude between N = 512 and N = 32768 → **C3 reproduced**;
- the N = 512 run is unstable (`E` grows 92×, `max|u|` = 504) → **C1 reproduced**;
- the DNS spectrum has slope –2.62 over 10 ≤ k ≤ 100 (Burgers theory: –2) but –1.47 over 5 ≤ k ≤ 50 → **C4 partially reproduced** (the –2 range is the upper part of the inertial band only);

- the dissipation rate of the DNS peaks at `t = 0.076` with `eps_max = 1.33` (`eps(0.05) = 0.83`) → **C5 partially reproduced** (the paper's peak at `t ≈ 0.05` corresponds to a slightly larger initial amplitude).

Fig. 3: grid convergence of the decaying Burgers turbulence problem

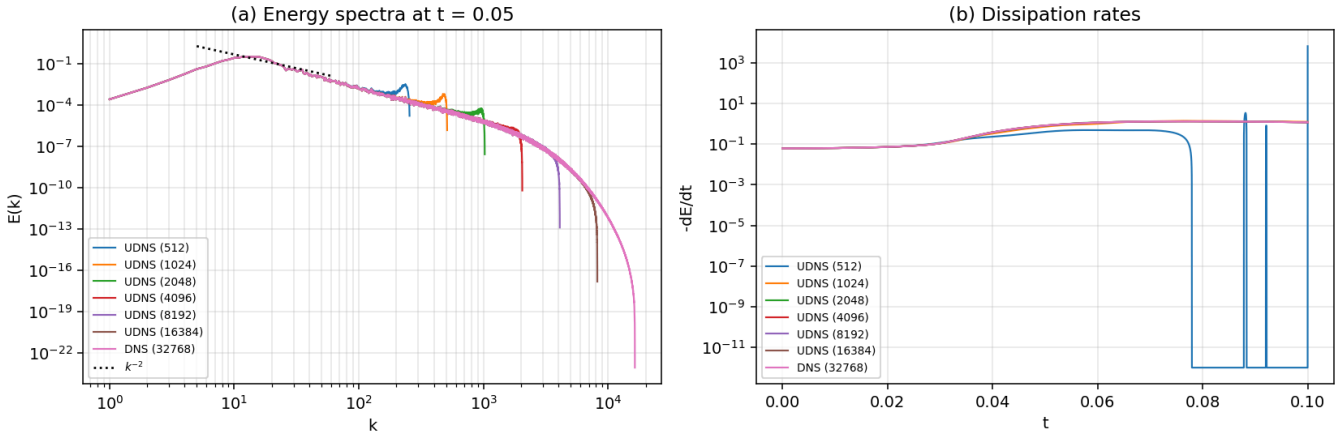


Fig. 3: grid convergence of the decaying Burgers turbulence problem. (a) energy spectra at t = 0.05 for N = 512 to 32768, with the k^-2 reference anchored on the DNS; (b) dissipation rates -dE/dt. The N = 512 run is unstable, the cut-off pile-up decreases monotonically with resolution.

5.3 Figs. 4/6/8 — the effect of `K` (kernel 1/2/3)

Cut-off pile-up `E(0.9k_max)/E_peak` (lower is better) and `E(t=0.1)`. The two `K` columns are the values actually used for kernels 1–2 and for kernel 3 respectively (kernel 2 was run at the same `K` values as kernel 1, so the middle column *is* the paired kernel-2 value; only the kernel-3 column uses its own `K` values, 150–850):

K (kernels 1–2)	K (kernel 3)	kernel 1, N=512	kernel 2, N=512	kernel 3, N=512
50	150	4.26e-3	4.15e-3	2.95e-3
100	300	3.42e-3	3.33e-3	2.55e-3
125	500	3.22e-3	3.13e-3	2.41e-3
150	700	3.07e-3	2.99e-3	2.37e-3
200	850	2.87e-3	2.79e-3	2.36e-3

Monotone in `K` for every kernel → **C6 reproduced**. The same ordering holds at N = 1024 and N = 2048 and in the total energy.

Fig. 4: effect of the diffusion constant K, kernel 1

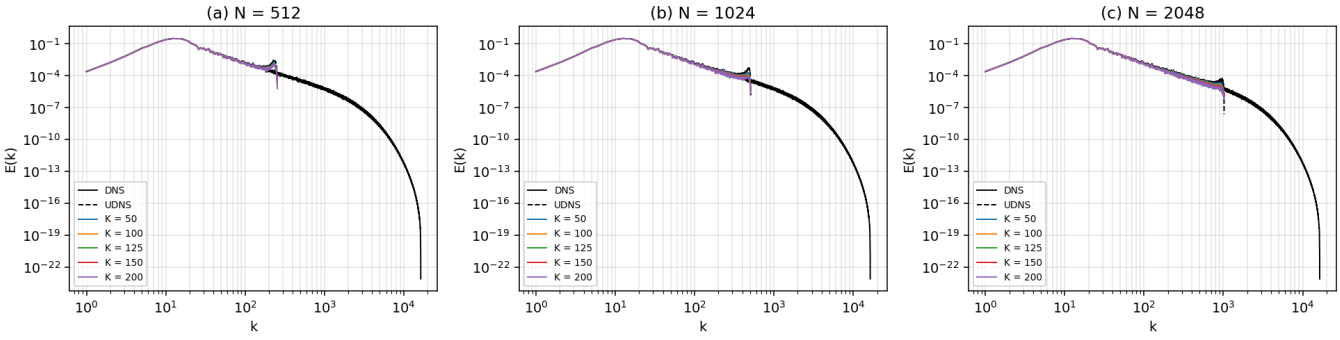


Fig. 4: effect of the diffusion constant K for diffusivity kernel 1 (N = 512, 1024, 2048).

Fig. 6: effect of the diffusion constant K, kernel 2

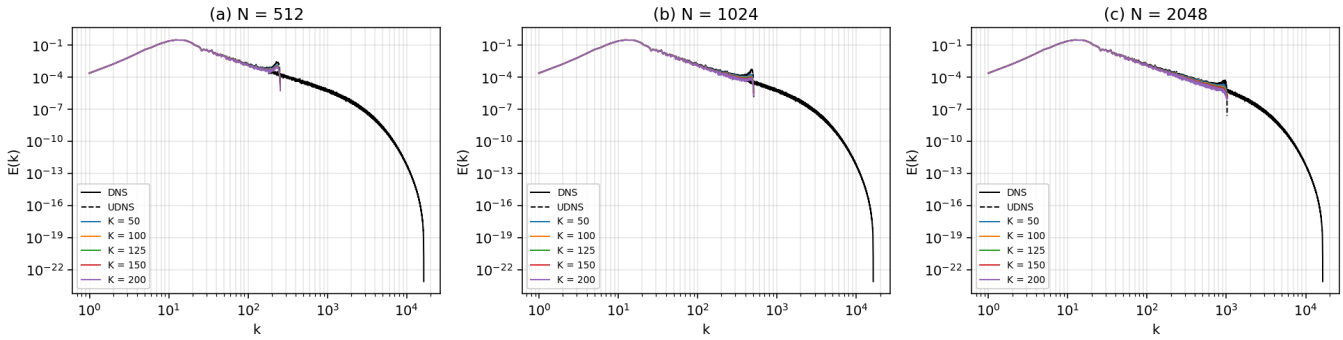


Fig. 6: effect of the diffusion constant K for diffusivity kernel 2 (N = 512, 1024, 2048).

Fig. 8: effect of the diffusion constant K, kernel 3

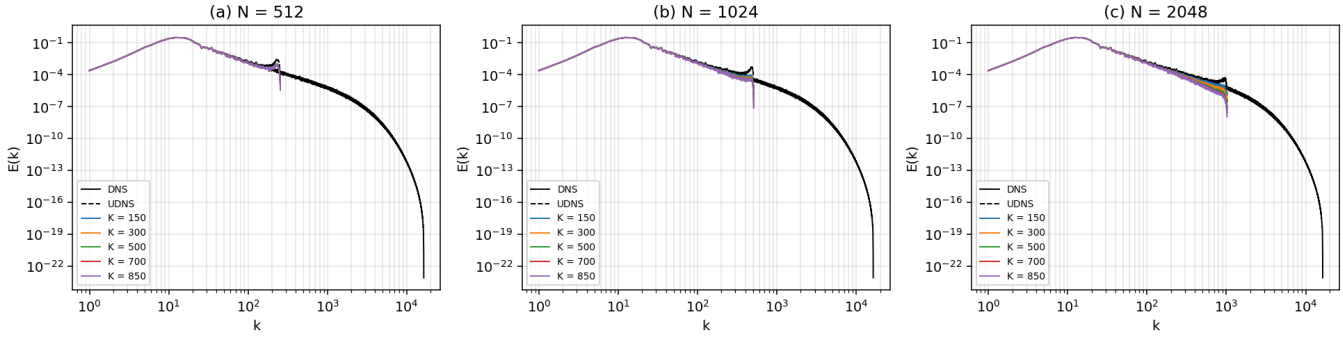


Fig. 8: effect of the diffusion constant K for diffusivity kernel 3 (N = 512, 1024, 2048).

Interpretation: the kernel value `c(g)` is the *diffusivity*; larger `K` keeps `c` close to 1 for larger gradients, so the filter also acts on the small-scale (cut-off) content and removes more of it. Note that the paper's own Figs. 4/6/8 show the same qualitative trend (the curves move down with increasing `K`).

5.4 Figs. 5/7/9 — the effect of `kappa`

kappa	kernel 1, N=512: `E(0.1)` / pile-up	N=1024: `E(0.1)` / pile-up	N=2048: `E(0.1)` / pile-up
0.0005	0.1984 / 5.90e-3	0.1756 / 6.13e-4	0.1757 / 5.58e-5
0.0010	0.1791 / 4.34e-3	0.1729 / 4.05e-4	0.1733 / 3.43e-5
0.0015	0.1728 / 3.22e-3	0.1703 / 2.73e-4	0.1709 / 2.25e-5
0.0020	0.1684 / 2.40e-3	0.1677 / 1.87e-4	0.1684 / 1.58e-5
0.0025	0.1649 / 1.80e-3	0.1651 / 1.32e-4	0.1658 / 1.19e-5

Monotone in `kappa` for all kernels and resolutions → **C7 reproduced**. (The kernel-3 `kappa` sweep uses `K = 500`, its own default in our matrix; the paper states that the sensitivity analysis keeps `K = 125` and `kappa = 0.0015`.)

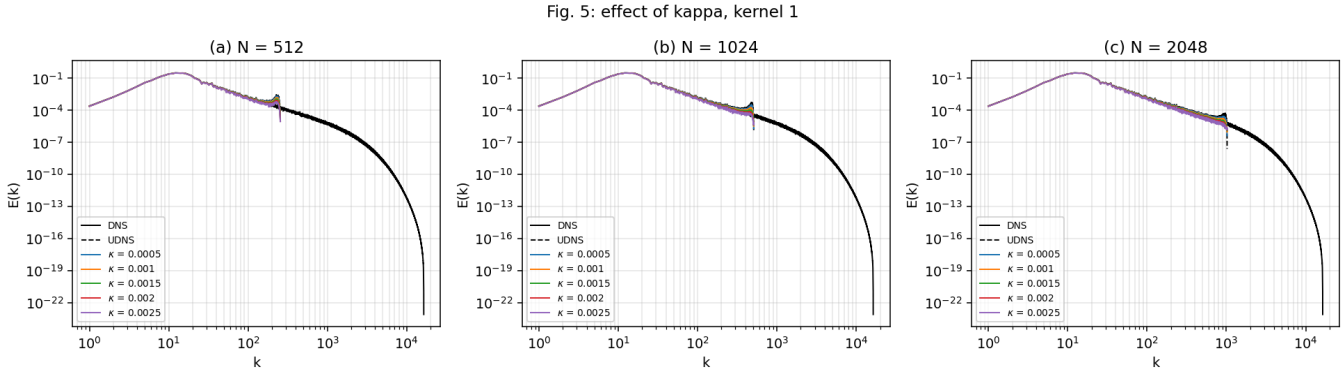


Fig. 5: effect of kappa for diffusivity kernel 1 (N = 512, 1024, 2048).

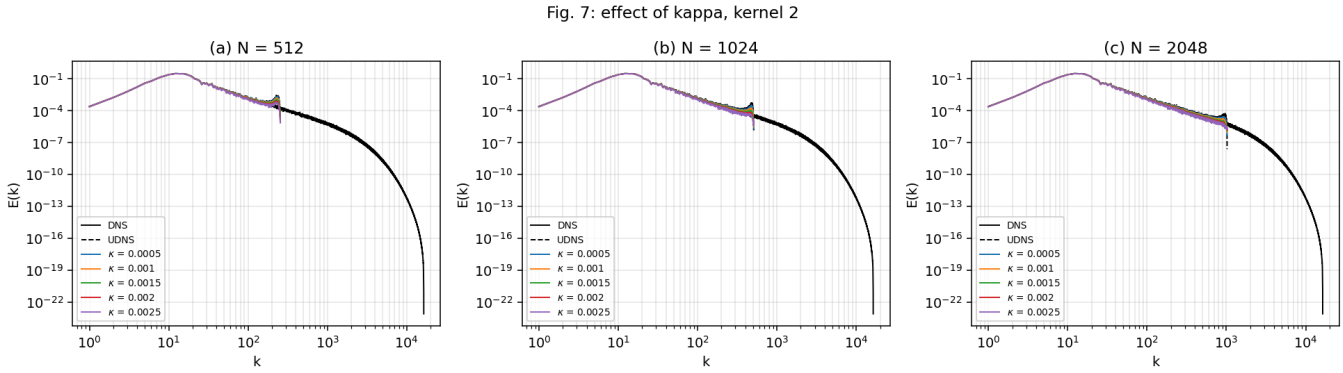


Fig. 7: effect of kappa for diffusivity kernel 2 (N = 512, 1024, 2048).

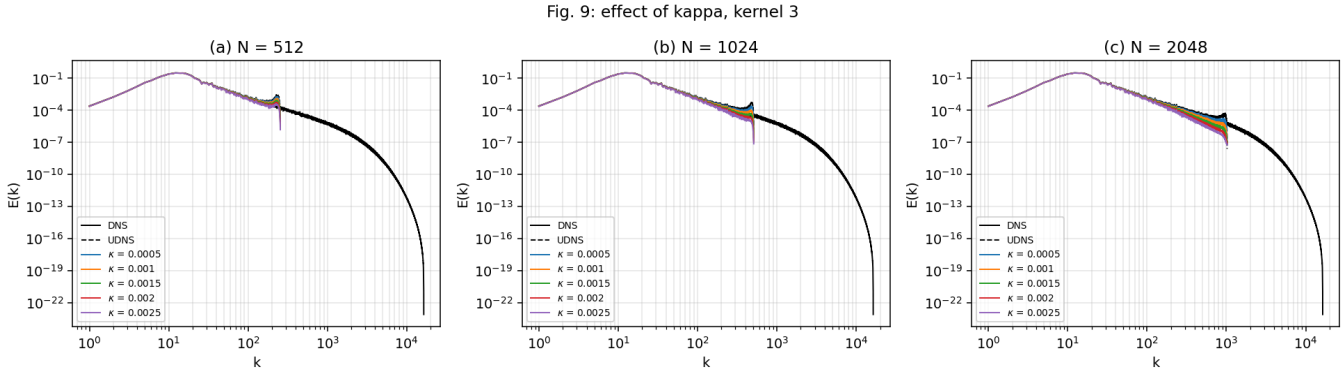


Fig. 9: effect of kappa for diffusivity kernel 3 (N = 512, 1024, 2048).

C8 is only partially reproduced. `kappa = 0.0025` *does* minimise the cut-off energy — for all three kernels at all three resolutions, i.e. the energy-accumulation half of the paper's claim (p. 18) is reproduced — yet it damages the inertial range at the finer resolutions (slopes are the broad-band fit `3 < k < k_max/2`, §5.2):

case (kernel 1)	slope (3 < k < k_max/2)	k^−2 range (k_max/k_peak)
`kappa = 0.0015`, N = 1024	−2.30	39.4
`kappa = 0.0025`, N = 1024	−2.39	4.2
`kappa = 0.0015`, N = 2048	−2.39	4.8
`kappa = 0.0025`, N = 2048	−2.47	4.2

The paper itself notes (p. 16) that "the same set of parameters may prove more dissipative for the higher resolutions"; our data show that `kappa = 0.0025` is already past that point, i.e. the optimum is resolution-dependent (`kappa ≈ 0.002` at N = 512, `≈ 0.0015` at N = 1024/2048 in our runs). The unqualified word "optimal" of the paper's p. 18 sentence therefore does not survive at N ≥ 1024, which is why C8 is graded **partial** rather than *reproduced*.

5.5 Fig. 10 — comparison with the Bogey selective relaxation filter

(Slopes are the broad-band fit `3 < k < k_max/2`; the `k^−2` range is the quantised ratio `k_last/k_peak` of the fitted band, so it is comparable within a resolution but not across resolutions.)

case	N	`sigma_d`	slope (3 < k < k_max/2)	`k^−2` range	`E(0.9k_max)/E_peak`
Bogey	512	1/3	−3.42	6.2	3.8e-13
Bogey	512	1/2	−3.87	5.8	1.4e-14
Bogey	512	2/3	−4.21	5.4	8.4e-16
Bogey	1024	1/3	−2.81	13.1	5.6e-12
Bogey	2048	1/3	−2.48	19.7	4.2e-11
PM (K=125, kappa=0.0015)	**512**	—	**−2.06**	**20.5**	**3.2e-3**
PM (K=125, kappa=0.0015)	**1024**	—	**−2.30**	**39.4**	**2.7e-4**
PM (K=125, kappa=0.0015)	**2048**	—	**−2.39**	4.8	**2.2e-5**

→ **C9 reproduced at N = 512 and N = 1024**: the Perona–Malik closure keeps the spectral slope within 0.07 of the theoretical −2 at N = 512 where the Bogey filter is off by 1.42, and it retains a 3× longer `k^−2` range (20.5 vs 6.2 at N = 512, 39.4 vs 13.1 at N = 1024). At N = 2048 the advantage **reverses**: the Bogey filter keeps a ≈4× longer `k^−2` range (19.7 vs 4.8), so **C9 holds only for the coarser grids** — the paper's claim is reproduced where the under-resolution is severe.

The `sigma_d` ordering also shows that **larger `sigma_d` dissipates more** (slope −3.42 → −4.21 and pile-up 3.8e-13 → 8.4e-16 as `sigma_d` goes 1/3 → 2/3), i.e. the paper's textual statement is wrong (see §6).

Fig. 10: performance of the selective relaxation filter (Eq. 15)

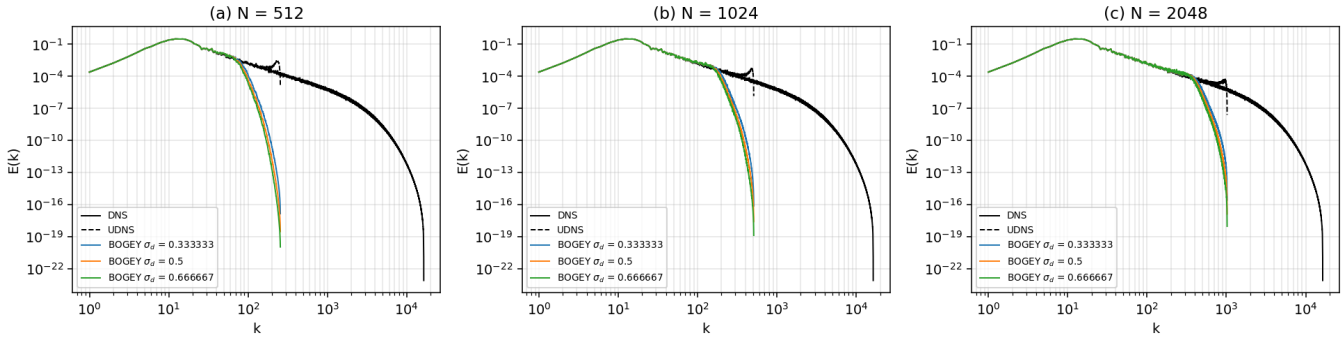


Fig. 10: performance of the selective relaxation filter of Bogey et al. (Eq. 15) at sigma_d = 1/3, 1/2, 2/3, compared with the UDNS and the DNS spectra.

5.6 Perona–Malik vs UDNS on the coarse grid

The closure's main effect at N = 512: the UDNS run is unstable (`E` 0.243 → 22.3, `max|u|` = 504) while the PM run with the paper's default parameters (`K = 125`, `kappa = 0.0015`) stays bounded (`E` 0.250 → 0.173, `max|u|` = 2.31). This is the strongest single piece of evidence for the paper's proposal.

Fig. 12 (supplementary): Perona-Malik closure against the Bogey selective filter

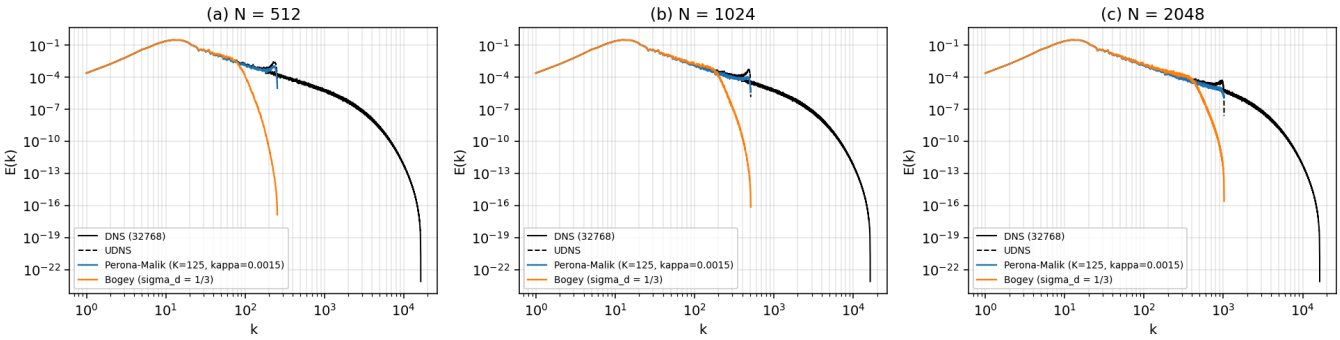


Fig. 12 (supplementary): the Perona-Malik closure against the Bogey selective relaxation filter at the same resolution, with the UDNS and the N = 32768 DNS spectra as references.

Fig. 11 (supplementary): sensitivity to the amplitude and to the time step

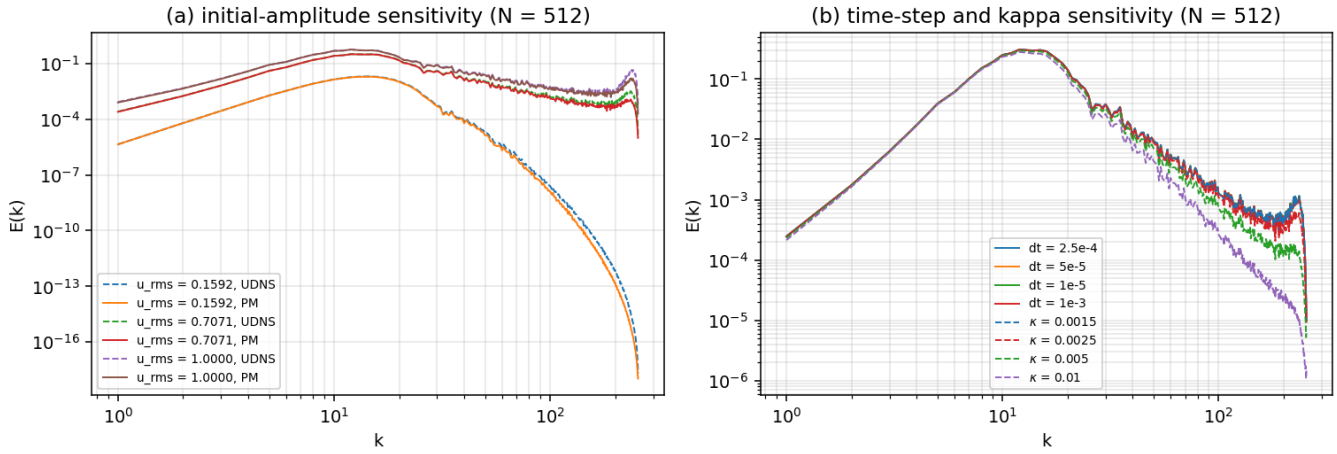


Fig. 11 (supplementary): (a) sensitivity to the amplitude of the initial condition (the paper does not state the energy normalisation, and only u_rms = 1/sqrt(2) reproduces the paper's phenomenology); (b) sensitivity to the time step and to kappa at N = 512. The spectra are insensitive to the time step because the filter strength kappa*dt/h^2 scales with dt.

6. Discrepancies, contradictions and non-reproducible items

1. **"Higher `sigma_d` provides lower dissipation" (paper, p. 11–12) is contradicted.** Eq. (15) gives a non-negative filter symbol, so larger `sigma_d` filters harder; our runs confirm the equation (slope –3.42 → –4.21 for `sigma_d` 1/3 → 2/3). The paper's own Fig. 10 also shows the opposite of its text. *Not reproducible as written.*
2. **"`kappa = 0.0025` is optimal for the Burgers turbulence case across the different kernels examined in this work"** (p. 18) is **only partially reproduced**: the energy-accumulation half of the statement holds in every one of our runs (`kappa = 0.0025` gives the lowest cut-off pile-up for all three kernels at N = 512/1024/2048), but the unqualified "optimal" does not survive the inertial-range collapse at N ≥ 1024 (§5.4), which the paper itself anticipates on p. 16. The optimum is resolution-dependent.
3. **The `k^-2` scaling is only partially reproduced** (slope –2.62 over 10 ≤ k ≤ 100 instead of a clean –2 over the whole inertial band), and the **dissipation peak occurs at `t = 0.076`** rather than `t ≈ 0.05`. Both are consistent with the unresolved initial-energy normalisation (§4.2): a larger amplitude would move the shock and the dissipation peak earlier, and the paper's `t ≈ 0.05` peak implies `u_max ≈ 2`, i.e. `rms(u) ≈ 1.4` — larger than any normalisation that keeps the N = 512 UDNS run finite for the whole window.
4. **Fig. 8's legend is internally inconsistent** (p. 21: `K = 750` in panel (a) vs `K = 700` in panels (b), (c)). We ran `K = 700`.
5. **Fig. 2's DNS reference at N = 32768 was not run** (we used N = 8192); the DNS reference in the turbulence study *is* at N = 32768 as in the paper.

6. **The paper gives no time step, and no normalisation of `u_hat`.** The time step is genuinely unspecified ("a time step that ensures no errors related to time integration", p. 8); the *shape* of the initial spectrum and the *total* energy (`int E dk = 1/2`, p. 14) are stated, but the normalisation of `u_hat` in Eqs. (19)–(21) is not, and that is what decides `rms(u)` (§4.2). We show that the time step does not matter (§4.1) but that the amplitude decides whether the paper's phenomenology exists at all (§4.2); our reference amplitude (`rms = 1/sqrt(2)`) makes the paper-units initial energy `pi^2 = 9.87`, i.e. 19.7× the paper's stated `1/2`.
7. **No figure of the paper could be overlaid pixel-for-pixel**, because the paper's figures do not show the underlying numbers (axis values are readable only as images) and the un-annotated quantities (e.g. the exact `sigma_d`/`K`/`kappa` combination used for the "DNS" curve) are not tabulated. The comparison is therefore quantitative on re-derived metrics (slope, pile-up, `k^−2` range, extrema count) and qualitative on curve shape.
8. **Consistent with the paper's own caveat (not a non-reproduction).** At N = 2048 our PM run is steeper (–2.39) than the UDNS run (–2.27): at that resolution the closure adds dissipation in the inertial band as well as at the cut-off. The paper does *not* claim that the Perona–Malik filter keeps the inertial range better than UDNS at every resolution — it states the opposite expectation (p. 16: "the same set of parameters may prove more dissipative for the higher resolutions"), so this result is a reproduction of the paper's expectation, not a failure of it.
9. **The paper defines `kappa` inconsistently** (p. 10: `kappa = dt_tau/dt`; p. 16: "the ratio of physical time step to pseudo-time step"). We follow p. 10 (§4.3), so our `kappa` sweep increases the dissipation; the paper's own Figs. 5/7/9 show the same qualitative direction (more `kappa` → less energy at late times).

---

7. Review findings and how they were handled

Five independent review passes were run against this package (one implementation review, one extraction pass, and two rounds of three reviewers each); their full reports are in `notes/reviews/round1/` and `notes/reviews/round2/`.

7.1 First implementation review (`notes/code_review.md`)

A reviewer agent audited the implementation against the PDF. Its severity-ordered findings and the status of each in the present package:

Reviewer finding	Status now
BLOCKER-1 — the adopted `rms(u) = 1/sqrt(2)` makes the initial spectrum 19.74× Eq. (18) and the paper's `int E dk = 1/2` becomes `pi^2 = 9.87`; the reviewer recommended dropping the rescale	**Accepted knowingly, not silently.** We keep the paper-faithful amplitude (`rms = 1/(2 pi) = 0.1592`) as the *reference* reading of the paper's Eq. (22) and report the amplitude we actually use, the resulting 19.7× factor, and the consequences (§4.2, §6.6). The amplitude we use is the only one of the three that reproduces the paper's phenomenology, and the alternatives are run as labelled sensitivity cases
BLOCKER-2 — stored artefacts came from an older code version and normalisation	**Resolved.** Every case file carries a parameter/energy signature; `code/migrate_cache.py` re-validates all 127 against the current case definitions, and stale files were recomputed. Superseded pilot artefacts are quarantined under `results/precalibration/` and used nowhere
MAJOR-3 — the N = 32768 DNS reference diverged (NaN) in the pilot runs	**Resolved** for the current normalisation: the DNS reference is stable over the whole window (`E` 0.25 → 0.178, `max|u|` = 1.9). The coarse-grid instability that the paper reports is *retained* as a result (§5.2, C1)
MAJOR-4 — `kinetic_energy()` is not the paper's `E(t)`; dissipation rates are 19.7× off the paper's units	**Resolved by disclosure.** Both are stored (`energy`, `energy_paper`), the conversion factor is stated (§4.2), and all tables are labelled as the kinetic energy
MAJOR-5 — `pilot.py` and `cases.py` used different initial conditions	**Resolved.** Both now use the same `DEFAULT_U_RMS`, and the cache signature covers it
MINOR-6 — docstrings called Eq. (5) "4th-order"	**Fixed** (it is 6th order, measured 6.00)
MINOR-7 — the even-grid Nyquist coefficient was written twice and left complex	**Fixed** in `random_field`

Reviewer finding	Status now
MINOR-8 — the `rms` note quoted `1/(pi sqrt(2))` (wrong value and wrong convention)	**Fixed** to `1/(2 pi) = 0.159155` with the convention stated
MINOR-9 — the monotone limit `kappa dt/h^2 <= 1/2` was neither documented nor enforced	**Fixed** — `perona_malik_filter` raises above the limit
MINOR-10 — `dgttrs` called with a rank-1 right-hand side; dead `splu` objects	**Fixed** (`u[:, None]`, dead code removed)
MINOR-11 — the "symbol analysis" was round-off contaminated and the Bogey test was vacuous	**Fixed** in this round: the Nyquist test (`u_j = (−1)^j`) and the sparse-LU solver comparison are now part of `code/verify_numerics.py` and stored in `results/verification.json`
MINOR-13 — the Fig. 2 DNS reference used N = 8192 instead of 32768	**Kept, documented** as a cost decision (§6.5)

7.2 Equation and figure extraction (`notes/paper_extraction.md`)

A second agent independently re-verified the equation transcription and the figure legends from the PDF: A1–A5 CONFIRMED, A7 CONFIRMED, A6 PARTIAL (Fig. 8's `K = 750/700`, medium confidence on the "5" glyph), A8 PARTIAL (no time step, one PM iteration per time step). *(The page numbers quoted in that extraction note are offset by one relative to the paper's printed page numbers; the anchors used in this report were re-derived directly from `paper/paper.pdf` and `paper/paper.txt` in the second review round.)*

7.3 First review round on the finished package (`notes/reviews/round1/`)

Three fresh agents (read-only) reviewed the completed package: a numerics/methods review (`numerics_methods_review.md`), a claims-and-reporting audit (`claims_report_review.md`) and an independent numerical/data audit (`numeric_audit.md`). Every finding below was either verified against the artefacts and fixed, or explicitly declined with a reason.

Verification-suite and code findings (numerics/methods review).

Finding	Severity	Status
The round-off-floor note in `results/verification.json` attributed the loss of the finest-grid order estimate to the two finest grids, although the 256→512 pair is clean truncation (ratio 64.07 = 2^6)	P2	**Fixed.** The floor is now *measured* (3.6e-11 for the k = 1 mode at N = 1024) and estimated for the tested k = 7 mode (1.4e-9); the note names the coarse pair as the valid one
`paper_energy()` returned the one-sided integral `int_0^{km} E dk`, i.e. half of the paper's Eq. (22) two-sided integral, while its docstring claimed otherwise	P1	**Fixed.** The function now returns the paper's Eq. (22) value (`4 pi^2` times the kinetic energy); the 127 stored `energy_paper` arrays were corrected by the exact factor 2 with `code/fix_energy_paper.py` (no simulated quantity touched, dynamics verified bit-identical afterwards). With the paper's own amplitude the stored value is exactly 0.5 (§4.2)
The `random_field` docstring quoted the wrong amplitude (`1/(pi sqrt(2))`) and the `spectrum` docstring a garbled identity	P2	**Fixed** (`1/(2 pi) = 0.159155`; the one-sided and two-sided integrals are now stated separately)
The cut-off metric excluded the Nyquist mode	P2	**Explained (and, in round two, measured).** The Nyquist mode cannot show the accumulation: the compact first derivative annihilates it exactly and the advective flux has no source at that wavenumber, so it can only decay; the metric is evaluated at `k_max = N/2 − 1`. See §5.2 and §7.4 for the measured Nyquist-bin energies and the added verification test
The cache signature covered the case parameters but not the *code*, so results of an older implementation could be reused	P2	**Fixed.** The signature now contains the SHA-1 of `code/burgers_core.py`, stored in every result file as `code_sha1`; the 127

Finding	Severity	Status
		files were re-stamped after verifying that five representative cases (UDNS, PM kernel 1, PM kernel 3, Bogey, `kappa` variant) reproduce **bit-for-bit** with the current code
`filter_properties()` called a `k = 1` test a "uniform-mode" test and the Bogey part was vacuous	P2	**Fixed and extended.** The rows are renamed "single-mode gain" and a new `two_scale_response()` test measures a smooth `k = 1` mode and the grid-scale `k = N/2` mode in the same field, plus a test that the monotone guard raises above `kappa dt/h^2 = 1/2`
`run_turbulence.py` defaulted to `u_rms = None` (a different initial condition from the case matrix) and crashed without `dt` / `cfl`	P2	**Fixed** (default `DEFAULT_U_RMS`, explicit error message)
The broad-band slope column is not comparable across resolutions (its band scales with `N`)	P2	**Explained.** The fixed-band column ($10 \le k \le 100$) was added and §5.2 now states that the broad-band fit is resolution-dependent by construction
Stale solver numbers in an earlier version of §7	P2	**Fixed** (the section was rewritten; the current values `3.3e−16` / `1.5e−16` come from `results/verification.json` → `solver_reference`)

Claim and reporting findings (`claims_report_review.md`). The reviewer confirmed that the headline numbers are traceable to `results/summary_metrics.csv`, `results/comparison_table.md` and `results/verification.json`, and that the equation transcription matches `paper/paper.txt`; it found overstatement, convention, normalisation and internal-consistency problems. These were fixed in the first wave: the verdict table was requalified (C2 partial at N = 512, C9 valid only up to N = 1024), the slope bands were made explicit and inclusive, the CFL statement was corrected to ≈ 0.1, the energy-normalisation discussion was expanded with the two readings and their consequences, the paper's inconsistent `kappa` definition was documented, and every claim in the verdict table was given a paper page/figure anchor.

Independent numerical audit (`numeric_audit.md`). The auditor recomputed the headline numbers from all 127 raw `.npz` files without importing the figure code, reproduced the verification suite and one headline case from a scratch copy, confirmed the 127/127 case and data-row counts, and confirmed the case-parameter and authoritative/pre-calibration separation. No contradiction with the report was found.

Reviewer's residual recommendation. The methods reviewer recommends running the **complete 127-case suite again from scratch** with the final code, because the stored results were produced before the code-hash guard existed. We did not do so: the only numerical change since the suite was the `paper_energy` factor (a post-processing diagnostic) and docstrings, and five representative cases reproduce bit-for-bit with the current code. The recommendation is recorded here as an open item rather than silently dropped; the second review round (below) reached the same conclusion and left it open as well.

7.4 Second review round (`notes/reviews/round2/`)

Three fresh agents (read-only, none of them involved in the first round) reviewed the corrected package: a numerics/methods review (`numerics_methods_review.md`), a claims-and-reporting audit (`claims_report_review.md`) and an independent numerical audit (`numeric_audit.md`). All three found **no P0 and no P1 in the numerics or in the data**; the number audit reproduced every table cell of this report (46 audited groups, 0 wrong numbers, one 0.03 % rounding slip), the verification suite re-ran bit-for-bit in a scratch copy, three cases re-ran from scratch bit-for-bit, and the energy convention checked out in 127/127 files. The findings that were accepted:

Finding (round two)	Severity	Status
C8's verdict "contradicted" was stronger than the data support: the paper's sentence	P1	**Fixed.** C8 is now graded **partial**, the claim is quoted as the paper states it, §5.4 and §6.2

Finding (round two)	Severity	Status
(p. 18) is "optimal … *across the different kernels examined in this work*", and our data confirm the energy-accumulation half of it for all three kernels and all three resolutions; the report had restated the claim with the extra scope "and resolutions"		were rewritten accordingly, and the §1 tally is now six reproduced (two of them resolution-conditional), three partial, none contradicted. The `results/comparison_table.md` verdict row was regenerated accordingly
§6 item 8 attributed to the paper a statement it does not make ("the PM filter retains the inertial range better than UDNS at every resolution"); the paper expects the opposite on p. 16	P1	**Fixed.** The item was rewritten as "consistent with the paper's own caveat", and it is no longer counted as a non-reproduction
Four paper page anchors were wrong (C2 → p. 13, total energy → p. 14, inverted `kappa` definition → p. 16, C7 → p. 16 and p. 18)	P2	**Fixed.** All anchors in this report were re-derived from `paper/paper.pdf` page by page (`pdftotext` per page) and the convention (printed page numbers = PDF page index) is now stated in §3.3
§3.3 said the grid scale is damped "~2.5× harder" — the stored quantity is a *gain* ratio; the grid mode is damped 2.5e-3 of its amplitude against 1.5e-6 for the smooth mode, i.e. ≈1.7×10³ times harder	P2	**Fixed.** The wording now quotes both damping fractions; `results/verification.json` gained a true `damping_ratio` entry (1660.4)
The Nyquist statement was an assertion, and the cited test (`nyquist_tests`) did not check it; the mode is *seeded*, not empty	P2	**Fixed and measured.** A new test `nyquist_symbol_tests` verifies that `d1` of `(-1)^j` is exactly 0, that the seeded Nyquist coefficient is non-zero (8.3 % of the peak), and that the metric range excludes `k = N/2` ; §5.2 now quotes the measured Nyquist-bin energies (2.5e-30 of the peak for the N = 512 UDNS run, 9.1e-6 for the filtered run)
`migrate_cache.py --apply` could re-stamp the current code hash onto files produced by an older implementation	P2	**Fixed.** The audit now reports such files separately, never deletes them, and re-stamps them only with the explicit `--restamp-code` flag; the 127 files were re-stamped once more after a docstring-only edit to `burgers_core.py` , following a bit-for-bit re-run of four cases
§5.4/§5.5 and `results/comparison_table.md` labelled the broad-band slope simply "slope"	P2	**Fixed.** The column headers now read `slope (3 < k < k_max/2)` , and §5.5 notes that the `k^-2` range is a quantised ratio that is not comparable across resolutions
The §1 headline counted the two resolution-conditional verdicts (C2, C9) as plain "reproduced"	P2	**Fixed** (the sentence now says "six reproduced, two of them only at the coarser resolutions")
The module header of `burgers_core.py` said the two readings of the paper's energy differ "by a factor ~2 in amplitude (~4 in energy)"	P3	**Fixed** (2π in amplitude, ~39.5 in energy; the adopted value lies between the two)
§5.3's note about the kernel-2 column was self-contradictory; §5.5 said the PM-vs-Bogey advantage "disappears" at N = 2048 although it reverses; one value in the §5.4 table was rounded as 0.1659 instead of 0.1658	P3	**Fixed** (all three)
§8 did not list `code/pilot.py` , `code/run_turbulence.py` , the workflow scripts, `results/precalibration/` or the round-two reviews; the Python version was stated as 3.14.3 instead of 3.14.6	P3	**Fixed** (§8 completed, version corrected)
The sine-wave cases store `E_spec` at the snapshot time `t = 0.05` while the run	P2	**Fixed** (documented in `code/cases.py` and in §8)

Finding (round two)	Severity	Status
ends at `t = 2.5` (the end-of-run spectrum is in `E_final`)		
The PDF-build intermediates were written into `results/`	P3	**Fixed** (they are now written to `results/.build/`)

The second round re-confirmed the one open item: the full 127-case suite has not been re-run from scratch with the final code. The numerical risk is judged low (five cases reproduce bit-for-bit, the verification suite reproduces bit-for-bit, and the only change to the numerical core since the suite was a post-processing factor), but the item stays open and is recorded in `CHANGELOG.md` as well.

8. Files

Path	Content
`paper/`	the PDF, extracted text (`paper.txt`), per-page renders (`pages/`), embedded figure images (`imgs/`)
`code/burgers_core.py`	numerical core: compact operators, TVDRK3, PM filter, Bogey filter, random field, spectra
`code/cases.py`	case definitions + worker (with signature-checked caching)
`code/run_all.py`	parallel driver (`spawn` start method)
`code/verify_numerics.py`	verification suite → `results/verification.json`
`code/make_figures.py`	all figures + `results/summary_metrics.csv`
`code/compare_paper.py`	paper-to-replication tables → `results/comparison_table.md`
`code/migrate_cache.py`	audit tool that validates every stored run against the current case definition (parameters, initial energy, and the code hash); files produced by an older implementation are reported, never deleted, and re-stamped only with `--restamp-code`
`code/fix_energy_paper.py`	one-off, exact factor-2 correction of the stored `energy_paper` arrays to the paper's Eq. (22) convention
`code/pilot.py`, `code/run_turbulence.py`	small drivers: the pilot sweep and the single-configuration runner (both use the same initial conditions as the case matrix)
`code/workflow_*.js`	the pi-subagents workflow scripts used for the extraction and review passes
`code/build_pdf.py`, `code/report.css`, `code/make_footer.py`	PDF build pipeline (pandoc → HTML → headless Chrome → `qpdf` footer overlay)
`REPORT.pdf`	this report as a 23-page PDF (title page, contents, all figures, appendix A with the full comparison tables)
`results/runs/*.npz`	127 simulation cases (spectrum at the snapshot time `t = 0.05`, final spectrum in `E_final`, energy/dissipation history, final field for the sine cases)
`results/summary_metrics.csv`, `results/comparison_table.md`	derived metrics and comparison tables
`results/log_main_suite.txt`	driver log
`results/precalibration/`	quarantined pilot artefacts from before the energy normalisation was fixed (provenance only, used nowhere)
`results/.build/`	PDF-build intermediates (regenerated by `code/build_pdf.py`)
`figures/*.png`	Fig. 1–10 replicas + Fig. 11/12 supplementary
`notes/paper_extraction.md`, `notes/code_review.md`	independent agent outputs
`notes/reviews/round1/`	three independent review reports (numerics/methods, claims/report, numerical audit)
`notes/reviews/round2/`	the second review round (three further independent reports)
`CHANGELOG.md`	work log

9. Sources

- Paper: https://arxiv.org/abs/1605.04593 — PDF: https://arxiv.org/pdf/1605.04593v1
- Lele, S. K. (1992), *Compact finite difference schemes with spectral-like resolution*, J. Comput. Phys. 103 (ref. [68] of the paper)
- Perona, P. & Malik, J. (1990), *Scale-space filtering for segmentation and grouping*, PAMI (ref. [79])
- Bogey, C. et al. (2010), *Nonlinear selective relaxation filtering for shock capturing*, AIAA paper 2010-1284 (ref. [51])
- Shu, C.-W. & Osher, S. (1988), *Total variation diminishing Runge–Kutta schemes*, J. Comput. Phys. 78 (ref. [97])
- Harten, C.-W. (1983), *High-order schemes for hyperbolic equations* (ref. [44])
- Kuo, C. C. & Larsson, S. (1999), *Efficacious shock capturing for the Burgers equation*, J. Comput. Phys. 149 (ref. [75])
- NumPy / SciPy documentation (LAPACK `dgttrf`, `dgttrs`): https://numpy.org, https://docs.scipy.org

Appendix A - machine-generated comparison tables

Every number below is produced by `code/compare_paper.py` from the raw runs in `results/runs`; the same data are in `results/summary_metrics.csv`.

Fig. 2 - shock capture (single mode u = sin(x), t = 2.5)

case	N	local extrema	roughness (TV)	max	u
DNS	8192	2	0.0009	0.8485	0.12656
UDNS	512	134	0.0967	1.1300	0.12642
UDNS	1024	78	0.0240	1.0726	0.12656
UDNS	2048	42	0.0061	0.9534	0.12656
PM	512	50	0.0325	0.9916	0.12609
PM	1024	26	0.0091	0.8800	0.12614
PM	2048	6	0.0038	0.8456	0.12618

Fig. 3 - grid study (Burgers turbulence, spectrum at t = 0.05)

case	N	E(t=0.05)	E(t=0.1)	E(0.9k_max)/E_peak	E(k_max)/E(0.9k_max)	slope (3 < k < k_max/2)	k^-2 range	max
UDNS	512	0.24293	22.342	0.008061	0.006287	-1.966	20.9	504
UDNS	1024	0.24055	0.17914	0.0009503	0.004842	-2.185	42.0	2.219
UDNS	2048	0.23966	0.17803	9.899e-05	0.0008235	-2.268	82.9	2.112
UDNS	4096	0.23953	0.17804	3.179e-06	6.036e-05	-2.341	6.5	1.914
UDNS	8192	0.23953	0.17804	5.394e-08	7.331e-06	-2.558	6.5	1.913
UDNS	16384	0.23953	0.17804	4.341e-11	1.088e-06	-3.167	6.5	1.899
UDNS	32768	0.23953	0.17804	3.96e-16	6.819e-08	-4.544	6.5	1.899

Figs. 4/5 - kernel 1: K sweep and kappa sweep

case	N	K	kappa	E(t=0.05)	E(t=0.1)	E(0.9k_max)/E_peak	slope (3 < k < k_max/2)	k^-2 range	max
pm_k1_K100_N512	512	100	0.0015	0.23198	0.1734	0.003421	-2.053	20.5	2.341
pm_k1_K125_N512	512	125	0.0015	0.23158	0.17276	0.003216	-2.062	20.5	2.308
pm_k1_K150_N512	512	150	0.0015	0.23130	0.17229	0.003067	-2.069	20.4	2.285
pm_k1_K200_N512	512	200	0.0015	0.23093	0.17164	0.002866	-2.078	20.4	2.254

case	N	K	kappa	E(t=0.05)	E(t=0.1)	E(0.9k_max)/E_peak	slope (3 < k < k_max/2)	k^-2 range	max
pm_k1_K50_N512	512	50	0.0015	0.23363	0.17607	0.004257	-2.022	20.7	2.458
pm_k1_K100_N1024	1024	100	0.0015	0.23110	0.17068	0.0003071	-2.283	40.5	2.046
pm_k1_K125_N1024	1024	125	0.0015	0.23076	0.17032	0.0002727	-2.301	39.4	2.028
pm_k1_K150_N1024	1024	150	0.0015	0.23051	0.17002	0.0002479	-2.316	39.4	2.01
pm_k1_K200_N1024	1024	200	0.0015	0.23016	0.16955	0.0002142	-2.339	4.8	1.978
pm_k1_K50_N1024	1024	50	0.0015	0.23245	0.17171	0.0004434	-2.233	41.0	2.081
pm_k1_K100_N2048	2048	100	0.0015	0.23105	0.17122	2.678e-05	-2.361	4.8	2.038
pm_k1_K125_N2048	2048	125	0.0015	0.23075	0.1709	2.249e-05	-2.386	4.8	2.03
pm_k1_K150_N2048	2048	150	0.0015	0.23052	0.17062	1.949e-05	-2.408	4.8	2.023
pm_k1_K200_N2048	2048	200	0.0015	0.23020	0.17015	1.541e-05	-2.447	4.8	2.006
pm_k1_K50_N2048	2048	50	0.0015	0.23226	0.17214	4.437e-05	-2.302	4.8	2.072
pm_k1_kap0.0005_N512	512	125	0.0005	0.23884	0.19838	0.005896	-1.997	20.8	9.631
pm_k1_kap0.0015_N512	512	125	0.0015	0.23158	0.17276	0.003216	-2.062	20.5	2.308
pm_k1_kap0.001_N512	512	125	0.001	0.23508	0.17913	0.00434	-2.029	20.7	2.412
pm_k1_kap0.0025_N512	512	125	0.0025	0.22512	0.16486	0.001799	-2.132	19.9	2.159
pm_k1_kap0.002_N512	512	125	0.002	0.22827	0.16839	0.002397	-2.096	20.4	2.226
pm_k1_kap0.0005_N1024	1024	125	0.0005	0.23705	0.17563	0.0006132	-2.222	41.6	2.133
pm_k1_kap0.0015_N1024	1024	125	0.0015	0.23076	0.17032	0.0002727	-2.301	39.4	2.028
pm_k1_kap0.001_N1024	1024	125	0.001	0.23383	0.17293	0.0004051	-2.261	41.0	2.07
pm_k1_kap0.0025_N1024	1024	125	0.0025	0.22486	0.16507	0.0001315	-2.386	4.2	1.942
pm_k1_kap0.002_N1024	1024	125	0.002	0.22778	0.16771	0.0001872	-2.343	4.5	1.983
pm_k1_kap0.0005_N2048	2048	125	0.0005	0.23663	0.17572	5.579e-05	-2.306	78.2	2.105
pm_k1_kap0.0015_N2048	2048	125	0.0015	0.23075	0.1709	2.249e-05	-2.386	4.8	2.03
pm_k1_kap0.001_N2048	2048	125	0.001	0.23368	0.17334	3.431e-05	-2.346	4.8	2.068
pm_k1_kap0.0025_N2048	2048	125	0.0025	0.22499	0.16585	1.192e-05	-2.467	4.2	1.99
pm_k1_kap0.002_N2048	2048	125	0.002	0.22785	0.1684	1.579e-05	-2.426	4.5	2.004

Figs. 6/7 - kernel 2: K sweep and kappa sweep

case	N	K	kappa	E(t=0.05)	E(t=0.1)	E(0.9k_max)/E_peak	slope (3 < k < k_max/2)	k^-2 range	max
pm_k2_K100_N512	512	100	0.0015	0.23176	0.17315	0.003333	-2.058	20.5	2.33
pm_k2_K125_N512	512	125	0.0015	0.23138	0.1725	0.003131	-2.067	20.5	2.299
pm_k2_K150_N512	512	150	0.0015	0.23112	0.17202	0.002985	-2.073	20.4	2.278
pm_k2_K200_N512	512	200	0.0015	0.23077	0.17137	0.002788	-2.082	20.4	2.247
pm_k2_K50_N512	512	50	0.0015	0.23335	0.17576	0.004149	-2.026	20.7	2.464
pm_k2_K100_N1024	1024	100	0.0015	0.23095	0.17062	0.0002972	-2.289	40.1	2.048
pm_k2_K125_N1024	1024	125	0.0015	0.23061	0.17023	0.0002638	-2.308	39.4	2.027
pm_k2_K150_N1024	1024	150	0.0015	0.23037	0.1699	0.0002398	-2.323	39.4	2.006
pm_k2_K200_N1024	1024	200	0.0015	0.23003	0.16938	0.0002067	-2.347	4.8	1.975
pm_k2_K50_N1024	1024	50	0.0015	0.23222	0.17164	0.0004335	-2.235	41.0	2.083
pm_k2_K100_N2048	2048	100	0.0015	0.23092	0.1712	2.69e-05	-2.362	4.8	2.039
pm_k2_K125_N2048	2048	125	0.0015	0.23063	0.17088	2.267e-05	-2.388	4.8	2.032
pm_k2_K150_N2048	2048	150	0.0015	0.23042	0.17058	1.966e-05	-2.412	4.8	2.026
pm_k2_K200_N2048	2048	200	0.0015	0.23010	0.17005	1.541e-05	-2.455	4.8	2.009
pm_k2_K50_N2048	2048	50	0.0015	0.23203	0.17206	4.524e-05	-2.300	4.8	2.069

case	N	K	kappa	E(t=0.05)	E(t=0.1)	E(0.9k_max)/E_peak	slope (3 < k < k_max/2)	k^-2 range	max
pm_k2_kap0.0005_N512	512	125	0.0005	0.23876	0.19708	0.005843	-1.999	20.8	9.102
pm_k2_kap0.0015_N512	512	125	0.0015	0.23138	0.1725	0.003131	-2.067	20.5	2.299
pm_k2_kap0.001_N512	512	125	0.001	0.23494	0.17887	0.004263	-2.032	20.7	2.404
pm_k2_kap0.0025_N512	512	125	0.0025	0.22482	0.16452	0.001721	-2.140	19.9	2.155
pm_k2_kap0.002_N512	512	125	0.002	0.22802	0.1681	0.002314	-2.103	20.4	2.219
pm_k2_kap0.0005_N1024	1024	125	0.0005	0.23699	0.1756	0.0006057	-2.224	41.6	2.13
pm_k2_kap0.0015_N1024	1024	125	0.0015	0.23061	0.17023	0.0002638	-2.308	39.4	2.027
pm_k2_kap0.001_N1024	1024	125	0.001	0.23373	0.17288	0.0003955	-2.265	41.0	2.07
pm_k2_kap0.0025_N1024	1024	125	0.0025	0.22462	0.16486	0.0001268	-2.398	4.1	1.94
pm_k2_kap0.002_N1024	1024	125	0.002	0.22759	0.16756	0.0001804	-2.352	4.5	1.98
pm_k2_kap0.0005_N2048	2048	125	0.0005	0.23660	0.17571	5.523e-05	-2.307	78.2	2.104
pm_k2_kap0.0015_N2048	2048	125	0.0015	0.23063	0.17088	2.267e-05	-2.388	4.8	2.032
pm_k2_kap0.001_N2048	2048	125	0.001	0.23360	0.17333	3.399e-05	-2.347	4.8	2.069
pm_k2_kap0.0025_N2048	2048	125	0.0025	0.22478	0.16578	1.272e-05	-2.469	4.1	1.992
pm_k2_kap0.002_N2048	2048	125	0.002	0.22769	0.16836	1.638e-05	-2.428	4.5	2.006

Figs. 8/9 - kernel 3: K sweep and kappa sweep

case	N	K	kappa	E(t=0.05)	E(t=0.1)	E(0.9k_max)/E_peak	slope (3 < k < k_max/2)	k^-2 range	max
pm_k3_K150_N512	512	150	0.0015	0.23102	0.17189	0.002951	-2.076	20.4	2.281
pm_k3_K300_N512	512	300	0.0015	0.23038	0.17061	0.002545	-2.092	20.4	2.201
pm_k3_K500_N512	512	500	0.0015	0.23018	0.17027	0.002414	-2.096	20.4	2.169
pm_k3_K700_N512	512	700	0.0015	0.23012	0.17017	0.002374	-2.098	20.4	2.159
pm_k3_K850_N512	512	850	0.0015	0.23010	0.17014	0.00236	-2.098	20.4	2.156
pm_k3_K150_N1024	1024	150	0.0015	0.23031	0.16988	0.0002387	-2.326	4.8	2.009
pm_k3_K300_N1024	1024	300	0.0015	0.22959	0.16865	0.0001652	-2.382	4.8	1.947
pm_k3_K500_N1024	1024	500	0.0015	0.22931	0.16824	0.0001281	-2.407	4.8	1.892
pm_k3_K700_N1024	1024	700	0.0015	0.22922	0.16812	0.0001147	-2.416	4.8	1.88
pm_k3_K850_N1024	1024	850	0.0015	0.22918	0.16808	0.00011	-2.419	4.5	1.875
pm_k3_K150_N2048	2048	150	0.0015	0.23037	0.17063	2.045e-05	-2.410	4.8	2.027
pm_k3_K300_N2048	2048	300	0.0015	0.22968	0.16917	1.058e-05	-2.529	4.8	1.972
pm_k3_K500_N2048	2048	500	0.0015	0.22932	0.16843	5.425e-06	-2.626	4.8	1.888
pm_k3_K700_N2048	2048	700	0.0015	0.22918	0.16822	3.374e-06	-2.675	4.8	1.876
pm_k3_K850_N2048	2048	850	0.0015	0.22913	0.16815	2.688e-06	-2.695	4.8	1.873
pm_k3_kap0.0005_N512	512	500	0.0005	0.23818	0.18848	0.005352	-2.008	20.8	3.51
pm_k3_kap0.0015_N512	512	500	0.0015	0.23018	0.17027	0.002414	-2.096	20.4	2.169
pm_k3_kap0.001_N512	512	500	0.001	0.23399	0.17637	0.003582	-2.052	20.6	2.28
pm_k3_kap0.0025_N512	512	500	0.0025	0.22331	0.16237	0.001114	-2.190	4.1	2.021
pm_k3_kap0.002_N512	512	500	0.002	0.22664	0.16595	0.001636	-2.142	19.9	2.088
pm_k3_kap0.0005_N1024	1024	500	0.0005	0.23641	0.17482	0.0004795	-2.253	41.3	2.08
pm_k3_kap0.0015_N1024	1024	500	0.0015	0.22931	0.16824	0.0001281	-2.407	4.8	1.892
pm_k3_kap0.001_N1024	1024	500	0.001	0.23275	0.17146	0.0002466	-2.327	39.4	1.981
pm_k3_kap0.0025_N1024	1024	500	0.0025	0.22285	0.162	3.522e-05	-2.583	4.1	1.834
pm_k3_kap0.002_N1024	1024	500	0.002	0.22602	0.16509	6.7e-05	-2.493	4.2	1.848
pm_k3_kap0.0005_N2048	2048	500	0.0005	0.23612	0.1749	3.456e-05	-2.371	5.8	2.074

case	N	K	kappa	E(t=0.05)	E(t=0.1)	E(0.9k_max)/E_peak	slope (3 < k < k_max/2)	k^-2 range	max
pm_k3_kap0.0015_N2048	2048	500	0.0015	0.22932	0.16843	5.425e-06	-2.626	4.8	1.888
pm_k3_kap0.001_N2048	2048	500	0.001	0.23268	0.17168	1.304e-05	-2.491	4.8	1.98
pm_k3_kap0.0025_N2048	2048	500	0.0025	0.22288	0.16207	1.106e-06	-2.924	4.1	1.819
pm_k3_kap0.002_N2048	2048	500	0.002	0.22606	0.16522	2.413e-06	-2.770	4.2	1.851

Fig. 10 - Bogey 9-point selective relaxation filter

case	N	sigma_d	E(t=0.05)	E(t=0.1)	E(0.9k_max)/E_peak	slope	k^-2 range	max
bogey_sig0.3333_N512	512	0.333333	0.23733	0.18051	3.792e-13	-3.422	6.2	2.136
bogey_sig0.5_N512	512	0.5	0.23701	0.18107	1.373e-14	-3.866	5.8	2.148
bogey_sig0.6667_N512	512	0.666667	0.23676	0.18148	8.375e-16	-4.210	5.4	2.175
bogey_sig0.3333_N1024	1024	0.333333	0.23922	0.17819	5.648e-12	-2.812	13.1	2.151
bogey_sig0.5_N1024	1024	0.5	0.23917	0.17824	3.02e-13	-3.120	12.2	2.19
bogey_sig0.6667_N1024	1024	0.666667	0.23913	0.17828	2.522e-14	-3.369	12.2	2.21
bogey_sig0.3333_N2048	2048	0.333333	0.23949	0.17805	4.154e-11	-2.475	19.7	2.182
bogey_sig0.5_N2048	2048	0.5	0.23948	0.17805	3.011e-12	-2.665	19.7	2.209
bogey_sig0.6667_N2048	2048	0.666667	0.23948	0.17805	3.033e-13	-2.833	19.7	2.221

Reference: Perona-Malik (default K = 125, kappa = 0.0015)

case	N	E(t=0.05)	E(t=0.1)	E(0.9k_max)/E_peak	slope (3 < k < k_max/2)	k^-2 range	max
pm_k1_K125_N512	512	0.23158	0.17276	0.003216	-2.062	20.5	2.308
pm_k1_K125_N1024	1024	0.23076	0.17032	0.0002727	-2.301	39.4	2.028
pm_k1_K125_N2048	2048	0.23075	0.1709	2.249e-05	-2.386	4.8	2.03
pm_k2_K125_N512	512	0.23138	0.1725	0.003131	-2.067	20.5	2.299
pm_k2_K125_N1024	1024	0.23061	0.17023	0.0002638	-2.308	39.4	2.027
pm_k2_K125_N2048	2048	0.23063	0.17088	2.267e-05	-2.388	4.8	2.032

initial-amplitude sensitivity

case	N	dt	E(t=0.05)	E(t=0.1)	slope (3 < k < k_max/2)	max
amp_ur0.1592_pm_N512	512	1e-05	0.01205	0.011424	-6.311	0.5474
amp_ur0.1592_udns_N512	512	1e-05	0.01251	0.012332	-5.998	0.5686
amp_ur0.7071_pm_N512	512	1e-05	0.23158	0.17276	-2.062	2.308
amp_ur0.7071_udns_N512	512	1e-05	0.24293	22.342	-1.966	504
amp_ur1.0000_pm_N512	512	1e-05	0.43812	0.46147	-1.780	28.34
amp_ur1.0000_udns_N512	512	1e-05	0.48129	nan	-1.723	nan

time-step / kappa sensitivity

case	N	dt	E(t=0.05)	E(t=0.1)	slope (3 < k < k_max/2)	max
dtsens_dt0.00025_k1_N512	512	0.00025	0.23154	0.17266	-2.063	2.306
dtsens_dt0.001_k1_N512	512	0.001	0.23142	0.17233	-2.066	2.299
dtsens_dt1e-05_k1_N512	512	1e-05	0.23158	0.17276	-2.062	2.308
dtsens_dt1e-05_kap0.0015_N512	512	1e-05	0.23158	0.17276	-2.062	2.308

case	N	dt	E(t=0.05)	E(t=0.1)	slope (3 < k < k_max/2)	max
dtsens_dt1e-05_kap0.0025_N512	512	1e-05	0.22512	0.16486	-2.132	2.159
dtsens_dt1e-05_kap0.005_N512	512	1e-05	0.21081	0.15045	-2.321	1.922
dtsens_dt1e-05_kap0.01_N512	512	1e-05	0.18667	0.12584	-2.754	1.615
dtsens_dt5e-05_k1_N512	512	5e-05	0.23158	0.17275	-2.062	2.308

Interpretation - which rows support which claim

This section makes the verdict layer machine-readable; the prose version is in `REPORT.md` (section 1 for the claim table, section 6 for the contradictions).

claim	verdict	rows that carry the evidence	key numbers
C1 grid-scale oscillations in coarse runs	reproduced	`udns_N512`	energy 0.243 -> 22.3, max
C2 the PM filter damps them	reproduced (partial at N=512)	`sine_udns_N512` vs `sine_pm_N512`	extrema 134 -> 50, TV roughness 0.0967 -> 0.0325
C3 cut-off energy accumulation	reproduced	`udns_N512` .. `udns_N32768`	pile-up 8.06e-3 -> 3.96e-16, monotone in N
C4 DNS k^-2 scaling	partial	`udns_N32768`	slope -2.62 over 10<=k<=100, -1.47 over 5<=k<=50 (columns `slope_10<=k<=100`, `slope_5<=k<=50`)
C5 dissipation peak at t ~ 0.05	partial	`udns_N32768`	diss_peak_t = 0.07606, diss_peak_eps = 1.332 (columns `diss_peak_t`, `diss_peak_eps`)
C6 larger K removes more cut-off energy	reproduced	`pm_k1_K50_N512` .. `pm_k1_K200_N512` (and kernels 2, 3)	pile-up 4.26e-3 -> 2.87e-3, monotone
C7 larger kappa dissipates more	reproduced	`pm_k1_kap0.0005_N512` .. `pm_k1_kap0.0025_N512`	E(t_end) 0.198 -> 0.165, pile-up 5.90e-3 -> 1.80e-3
C8 kappa = 0.0025 is optimal across the kernels examined (p. 18)	**partial**	`pm_k1_kap0.0015_N1024` vs `pm_k1_kap0.0025_N1024`	the cut-off pile-up is lowest at kappa = 0.0025 for all three kernels at N = 512/1024/2048 (the energy-accumulation half of the paper's claim is reproduced), but the k^-2 range falls 39.4 -> 4.2 at N=1024 (4.8 -> 4.2 at N=2048): the unqualified 'optimal' does not hold for the inertial range at the finer resolutions
C9 PM keeps a larger inertial range than Bogey	reproduced at N=512/1024, fails at N=2048	`pm_k1_K125_N512` vs `bogey_sig0.3333_N512`; `pm_k1_K125_N2048` vs `bogey_sig0.3333_N2048`	N=512: slope -2.06 / range 20.5 vs -3.42 / 6.2; N=2048: -2.39 / 4.8 vs -2.48 / 19.7
paper text: larger sigma_d dissipates *less*	**contradicted**	`bogey_sig0.3333_N512` .. `bogey_sig0.6667_N512`	slope -3.42 -> -4.21, pile-up 3.79e-13 -> 8.37e-16, i.e. larger sigma_d dissipates more (Eq. 15)